

ISHAN BIJUKUCHHAY

Kitchener, ON ◇ (519)-497-9248 ◇ ishanbijukuchhay@gmail.com ◇ www.linkedin.com/in/ishan-bijukuchhay/

EDUCATION

Queen's University - Bachelor of Applied Science, Computer Engineering September 2025 - April 2030

PROJECT EXPERIENCE

HTTP Caching Reverse Proxy

2026

Personal Project

- Built an HTTP caching reverse proxy in C on a Raspberry Pi 5 (hash table with separate chaining, $O(1)$ lookup; doubly-linked LRU list, $O(1)$ eviction).
- Benchmarked on hardware with **wrk**: $\sim 46k$ req/s on cache hits vs $\sim 1.6k$ on origin fetches ($\sim 29\times$ throughput).
- Profiled concurrency tradeoffs and eviction policies (LRU held 75% hit rate vs LFU's 9.5% under drifting load).
- Hardened against allocation-failure, socket timeouts, and truncation-based cache poisoning.

Clean-Sweeping Robot

September 2025 - Present

Queen's VEX Robotics Club

- Collaborated on a team to design and build an autonomous cleaning robot on Arduino, programmed in Rust/C++.
- Designed and tested circuit connections for reliable sensor communication with the Arduino board.

EcoTrack - Carbon Footprint Tracker

2026

Team Lead, APSC 103

- Led a team of 5 to ship a full-stack web app (React, Flask) converting everyday choices into CO₂e estimates.
- Managed deadlines and client communication, delivering a working prototype that earned positive client feedback.

WORK EXPERIENCE

Lifeguard/Instructor

June 2025 - September 2025

City of Kitchener - Aquatics Division

- Planned, delivered, and adapted over 20 swim programs for 100+ participants across age groups.
- Tracked attendance, progress, and evaluation data for each student with Word and Excel.
- Managed tasks and communicated with team to ensure timely deliverables and mitigate risks.

EXTRACURRICULAR EXPERIENCE

General Manager

September 2022 - June 2025

Huron Heights S.S. Robotics Team

- Led and coordinated 15+ project teams building battle bots, managing schedules to ensure on-time delivery.
- Categorized members by skill and collaboration dynamics in Excel; presented competition rules and logistics via PowerPoint.

Public Relations Manager

September 2024 - June 2025

Huron Heights S.S. - Asian Students Association

- Coordinated planning of large-scale cultural events with local businesses, aligning student leaders and faculty sponsors.
- Tracked outreach, vendor contacts, and budget progress in Excel to maintain visibility on event milestones.

Mental Health Steward

2025 - Present

EngWell - Queen's Engineering

- Selected to provide peer-level mental health support and connect students to wellbeing resources.

SKILLS AND INTERESTS

Languages

C, C++, Rust, Python

Systems

Data Structures & Algorithms, Sockets, Embedded/Arduino, Linux

Hardware

PCB Design (Proteus), Fusion 360

Tools

Git, MS Excel, Word, Project, PowerPoint